

Physics A (H156, H556)

Exploring - Circuit symbols ppq

KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is 31.
- The total number of marks may take into account some 'either/or' question choices.

- 1(a) In an experiment a circuit is set up so that a capacitor with a resistor in series can be charged and at some later time discharged through the same resistor without changing the positions of the components. This process can be repeated.

The supply has a potential difference (p.d.) 6.0 V d.c.

The capacitor has capacitance 1.0 μF .

The resistor has resistance 10 k Ω .

A voltmeter is used to measure the p.d. across the capacitor.

Draw a circuit diagram for this experiment.

[2]

- (b) Calculate the charge Q stored on the capacitor when it is fully charged.

$Q = \dots\dots\dots\text{C}$ [1]

- (c) Use a calculation to explain why it will not be possible to measure the variation of p.d. across the capacitor with time, using a stop watch.

----- [4]

- (d) State how this experiment can be modified to measure the variation of p.d. across the capacitor with time as the capacitor charges.

----- [1]

- (e) The capacitor was completely charged and then discharged to 4.12 V.

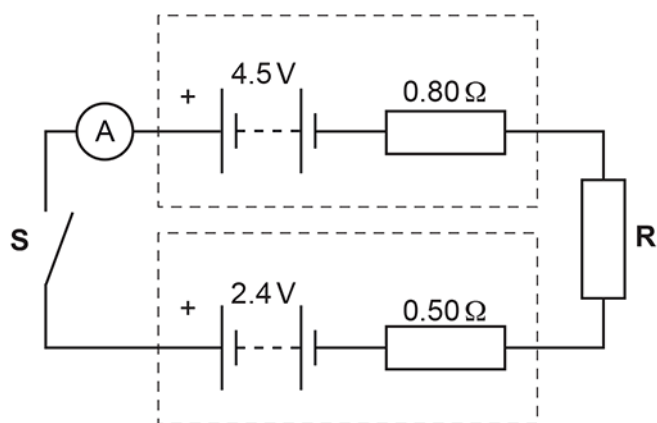
- i. Calculate the time t required for the p.d. across the capacitor to reach 4.12 V when discharging.

$t = \dots\dots\dots$ s [2]

- ii. Calculate the average rate at which energy is lost by the capacitor as it discharges from 6.0 V to 4.12 V.

average rate at which energy is lost = $\dots\dots\dots$ J s⁻¹ [3]

2 The circuit diagram of an electrical circuit is shown below.



The positive terminals of the batteries are connected together.

One battery has electromotive force (e.m.f.) 4.5 V and internal resistance $0.80\ \Omega$.

The other battery has e.m.f. 2.4 V and internal resistance $0.50\ \Omega$.

R is a coil of insulated wire of resistance $1.2\ \Omega$ at room temperature.

The switch **S** is closed.

- i. On the diagram, draw an arrow to show the direction of the conventional current.

[1]

- ii. Calculate the current I shown by the ammeter.

$I = \dots\dots\dots$ A [3]

- iii. The insulated wire has diameter $4.6 \times 10^{-4}\text{m}$.

The number density of charge carriers in **R** is $4.2 \times 10^{28}\text{ m}^{-3}$

Calculate the mean drift velocity v of the charge carriers in **R**.

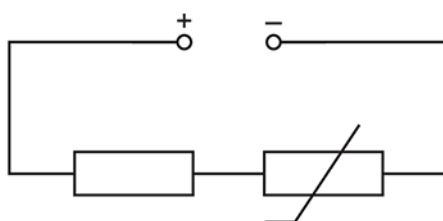
$$v = \dots\dots\dots \text{ m s}^{-1} \text{ [2]}$$

- iv. The current measured by the ammeter is smaller than that calculated in (ii). This is because the temperature of R increased due to heating by the current.

Without any changes to the circuit itself, state and explain what practically can be done to make the measured current the same as the calculated current..

----- [2]

- 3 A circuit with a thermistor is shown below.



The resistance of the resistor is R and the resistance of the thermistor is $2.5R$.
The potential difference (p.d.) across the thermistor is 5.0 V .

What is the total p.d. across both components?

- A 2.0 V
- B 7.0 V
- C 12.5 V
- D 17.5 V

Your answer

[1]

- 4 A filament lamp **X** is part of an electrical circuit. The circuit has a battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance. The potential difference across the lamp can be increased **continuously** from 0 to 6.0 V. This potential difference is measured using a voltmeter.

The lamp glows brightly at 6.0 V.

- i. Draw a circuit diagram for this electrical arrangement.

[2]

- ii. Describe and explain the variation of the resistance of this lamp as the potential difference across it is changed from 0 to 6.0 V.

[4]

- iii. The filament lamp **X** is now connected in a different circuit as shown in Fig. 16.

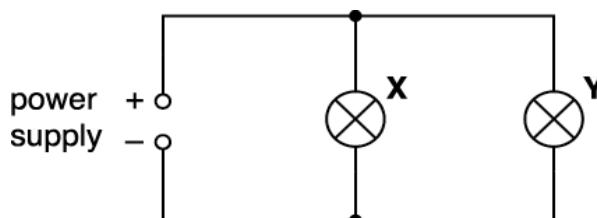


Fig. 16

The power dissipated in **X** is three times more than the power dissipated in the filament lamp **Y**. The filament wire of lamp **X** has a diameter half that of lamp **Y**.

The filament wires of **X** and **Y** are made of the same material and are at the same temperature.

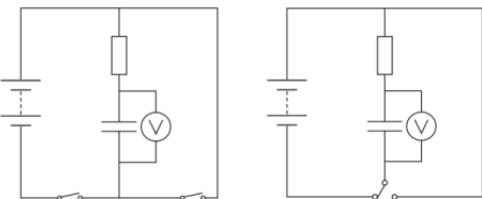
Calculate the ratio

$$\frac{\text{mean drift velocity of charge carriers in lamp X}}{\text{mean drift velocity of charge carriers in lamp Y}}.$$

ratio = [3]

END OF QUESTION PAPER

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
1	a	Circuit showing (6V) supply in series with a capacitor and resistor, with a voltmeter in parallel with the capacitor. Switch/switches allowing discharging of the capacitor through the resistor.	B1 B1	No labels required. ALLOW any suitable symbol for d.c. supply ALLOW this mark if resistor and capacitor in parallel if switch will allow the discharge Examples of correct circuit for both marks e.g.  Examiner's Comments Many candidates were able to correctly draw the supply, capacitor, and resistor in series with a voltmeter in parallel for a single mark. The position of a switch/switches to allow a discharge was less simple and many candidates had a single switch in series with their components. There are several ways to correctly draw this, and examiners allowed the use of double throw type switches as long as the idea was clear. Placing the capacitor and resistor in parallel with the supply may have allowed access to the second marking point as it would allow the discharge but not the charging. As always, clarity of the diagram makes it easier for examiners to understand. Several candidates appeared not to know the symbol for the capacitor and used a capital C in a circle. Less than one fifth of candidates were able to correctly draw the required diagram.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	Charge = $1.0 \times 10^{-6} \times 6.0 = 6.0 \times 10^{-6}$ (C)	A1	ALLOW correct answer to 1 significant figure <u>Examiner's Comments</u> The vast majority of candidates were able to calculate this correctly and the main reason for not awarding the mark tended to be from a power of ten error.
	c	(CR =) $1 \times 10^{-6} \times 10 \times 10^3$ CR / Time constant / $\tau = 0.010$ (s) Time constant / τ is time taken to fall to 1/e (37%) of initial value The time it takes to record the variation of pd / the capacitor to discharge is far less than (human) <u>reaction time</u> OR $V = V_0 e^{-\frac{t}{CR}}$ to give $0.6 = 6.0 e^{-\frac{t}{1.0 \times 10^{-6} \times 10 \times 10^3}}$ $t = 0.023$ (s) 0.6 is the voltage when it has fallen to 10% of the initial voltage The time it takes to record the variation of pd / for the capacitor to discharge is far less than human <u>reaction time</u> OR $V = V_0 e^{-\frac{t}{CR}}$ to give $V = 6.0 e^{-\frac{0.02}{1.0 \times 10^{-6} \times 10 \times 10^3}}$ $V = 0.81$ V 0.02 s is a very short amount of time The time it takes to record the variation of pd / for the capacitor to discharge is far less than human <u>reaction time</u>	C1 A1 B1 B1 (C1) (A1) (B1) (B1) (C1) (A1) (B1) (B1)	ALLOW POT error for capacitance if same as in (b) ALLOW 1sf ALLOW t for τ NOT fallen <u>by</u> 37% Substitution into exponential decay equation to fall to a value of V less than $0.9V_0$ (<5.4V). ALLOW calculation in log form e.g $\ln 0.6 = \ln 6.0 - t/0.01$ Justification/explanation for using a given voltage Substitution into exponential decay equation with a time less than 0.1s ($t < 0.1$s). ALLOW calculation in log form $\ln V = \ln 6.0 - 0.02/0.01$ Justification/explanation for using a given time <u>Examiner's Comments</u> Candidates were credited with calculations that produced a time or voltage that showed the time for decay was rapid. There were many possible routes to an answer and candidates were credited with any method that would produce a correct solution. As the question also included an explanation, it was often necessary to justify candidates' values for further credit rather than simply determine a numerical answer. The calculations were, in general,

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					completed well and a good number of candidates were able to score 2 or more marks. Explanations in terms of reaction time were quite rare and often a vague answer such as 'the time is short' or 'human error' was given. Many candidates also related it to the precision of the stopwatch rather than the limitations of their use due to human reaction time. Exemplar 2 the capacitor with time, using a stop watch. $V = V_0 e^{-\frac{t}{RC}}$ $1 = 6 e^{-\frac{t}{(6 \times 10^{-3}) \times (1 \times 10^{-6})}}$ $\ln\left(\frac{1}{6}\right) = -\frac{t}{(6 \times 10^{-3}) \times (1 \times 10^{-6})}$ $t = 0.0179 \text{ s}$ For the capacitor to discharge to 1V it takes less than a millisecond, which is less than human reaction time. [4] hence a large uncertainty due to random error In this response the candidate has chosen to determine the time taken to fall to 1 V. The calculation has been done correctly for the first 2 marks (although there is no unit, it is clear what t is). The third marking point has not been satisfied – a comment would be needed (however brief) on why 1 V was chosen. The final marking point is given for the idea that this is time is less than human reaction time.
	d		Use an oscilloscope / data logger with a voltmeter probe	B1	ALLOW connect a voltmeter to a datalogger Examiner's Comments While changing the value of the resistor or capacitor would increase the charging time, that would alter 'this experiment' and as such was not credited. In the context of this question, it is how the variation can be determined using the given values. Few candidates appreciated this, however there were significant numbers who knew that an oscilloscope could be used in place of the voltmeter.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	e	i	$V = V_0 e^{-\left(\frac{t}{CR}\right)} \text{ to give } 4.12 = 6.0 e^{-\left(\frac{t}{1.0 \times 10^{-6} \times 10 \times 10^3}\right)}$ $= 3.76 \times 10^{-3} \text{ (s)}$	C1 A1	ALLOW in terms of logs eg $\ln 4.12 = \ln 6.0 - \frac{t}{0.01}$ Correct to at least 2 significant figures IGNORE minus sign in final answer <u>Examiner's Comments</u> Most candidates were able to calculate the correct time for the discharge, by taking substituting values into the logarithmic equation. Common errors included incorrect taking of logs by division rather than subtraction (depending on their original set-up) and power of ten errors or transcription errors in the values of C or R. As with many calculations, those who spent a little time setting out the working carefully were more likely to get the correct answer.
		ii	Change in energy = $\frac{1}{2} CV_1^2 - \frac{1}{2} CV_2^2$ $= \left(\frac{1}{2} \times 1 \times 10^{-6} \times 6^2\right) - \left(\frac{1}{2} \times 1 \times 10^{-6} \times 4.12^2\right) = 9.5 \times 10^{-6} \text{ (J)}$ $\text{rate} = 9.5 \times 10^{-6} / 3.76 \times 10^{-3}$ $= 2.53 \times 10^{-3} \text{ (Js}^{-1}\text{)}$	C1 C1 A1	ALLOW POT error from (e)(i) Ecf from (e)(i) Correct to at least 2 significant figures <u>Examiner's Comments</u> Only around one third of candidates were able to correctly calculate this answer. By far the most common error was to calculate the change in energy using the difference in voltages (using 1.88 V) rather than calculating the separate energies and then subtracting. Several candidates left their answer at this point, rather than going on to divide by their value for the time.
			Total	13	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2		i	Arrow in anticlockwise direction	B1	Allow this mark for correct direction shown on diagram either on or off connecting wires <u>Examiner's Comments</u> This question required the candidates to appreciate that the sum of the emfs will lead to an anticlockwise conventional current. This question was answered well by the majority of candidates, however a number put two directions on, one from each cell.  Misconception The unusual setting out of the circuit meant that some candidates were unsure whether parts of the circuit were in series or parallel. This could have been overcome by following the circuit or even by redrawing it.
		ii	$(E =) 4.5 - 2.4$ or $(R_T =) 0.80 + 0.50 + 1.2$ $4.5 - 2.4 = I \times (0.80 + 0.50 + 1.2)$ $I = 0.84 \text{ (A)}$	C1 C1 A1	$E = 2.1 \text{ (V)}$; $R_T = 2.5 \text{ (}\Omega\text{)}$ Treat missing 1.2 resistance as TE Allow 2 marks for 2.8 (A); $E = 6.9 \text{ V}$ used <u>Examiner's Comments</u> This calculation required the candidate to set out the whole circuit in one. Around one third did not score any marks on this question as they attempted to treat each cell individually and then produce some form of average. Other common misunderstandings included treating the 0.5 ohm and 0.8 ohm resistors as if they were in parallel, and adding the emfs.

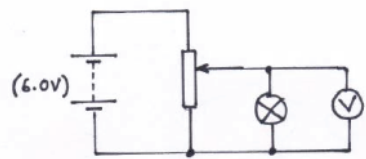
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	$(I = Anev)$ $0.84 = \pi \times (2.3 \times 10^{-4})^2 \times 4.2 \times 10^{28} \times 1.60 \times 10^{-19} \times v$ $v = 7.5 \times 10^{-4} \text{ (m s}^{-1}\text{)}$	C1 A1	Possible ECF from (ii) Note answer is $2.5 \times 10^{-3} \text{ (m s}^{-1}\text{)}$ for $I = 2.76 \text{ (A)}$ Allow 1 mark for 1.9×10^{-4} ; diameter used as radius <u>Examiner's Comments</u> This question was well done by a large number of candidates, many of whom scored full marks by correctly following through with their value of current from the previous part. Few candidates used the diameter instead of the radius when determining the cross sectional area, and for the most part the setting out of the calculation meant that credit could be given even if arithmetic errors occurred later.
		iv	Sensible suggestion, e.g. use a water bath / fan / only switch on when taking readings Need to lower the temperature / reduce resistance of R	M1 A1	Allow keep the surroundings cold Allow to keep the temperature / resistance constant OR allow increase in temperature increases resistance <u>Examiner's Comments</u> Candidates were expected to provide any method of cooling the circuit, or preventing it heating in the first place. A wide variety of solutions were given and as long it is viable, it was credited.  Misconception Some candidates gave perfectly viable solutions, but these involved changes to the circuit, which was not allowed in the question. It is very important to make sure than any response does fit what is being asked.
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
3			B	1	<u>Examiner's Comments</u> The correct response is B. This question was correctly answered by around two thirds of candidates. There appeared to be various routes to the correct solution; many opted to work out a current in terms of R, but the more elegant working was in terms of simple ratios which demonstrated a good understanding of p.d. in a series circuit. Encouragingly, very few candidates opted for response A, which was a p.d. below that of the thermistor alone. It should be noted that a couple of candidates put a '7' in the answer box – as correct working had been shown by them, and leading to the correct numerical value this was credited by examiners. However, this cannot be guaranteed to occur in other cases and candidates are to be encouraged to put only the correct letter.
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4		i	Correct circuit with a battery, potential divider, lamp and voltmeter. 	B1	
		i	Correct symbols used for all components.	B1	
		ii	Description: The temperature of the filament increases. (AW)	B1	Allow 'when cold the resistance is small'
		ii	The resistance of the lamp increases	M1	
		ii	from a non-zero value of resistance.	A1	
		ii	Explanation: Resistance increases because electrons/charge carriers make frequent collisions with ions. (AW)	B1	
		iii	($P = VI$) current in X is 3 times the current in Y Or area of X is 4 times smaller than area of Y	C1	Allow other correct methods.
		iii	$I = \frac{P}{V}$ and ratio = $\frac{3}{0.25}$	C1	
		iii	ratio = 12	A1	
Total			9		